

Data Normalisation

Machine Learning Lecture Slides

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Data standardisation

- For multivariate data, each variable may have different range
- e.g. Age (0-120), Dose (0-10), Cell size (0-0.1)
- ML algorithms may be **susceptible** to large variations in the range of different variables
- Data standardisation is a technique to eliminate such variations
- Before applying a ML algorithm, we need to check the assumptions it imposes on input data

Mean subtraction

- Transforms each variable to have mean 0 (aka *location transform*)
- For each variable x_i (excluding x_0):
 - replace with $(x_i - \mu_i)$
- Also applied to output variable Y
- Reversible operation (may not be relevant)

Original Data			
x_0	x_1	x_2	y
1	4	2	1.8
1	16	4	2.1
1	32	6	6.1
1	64	8	7.8
Sum	116	20	
Mean	29	5	

0 Mean Data			
x_0	$x_1 - \mu_1$	$x_2 - \mu_2$	y
1	-25	-3	1.8
1	-13	-1	2.1
1	3	1	6.1
1	35	3	7.8
Sum	0	0	
Mean	0	0	

Variance normalisation

- Transforms each variable in terms of distance from mean with distance measured in units of **s.d.** (aka *scale transform*)
- Transforms data to **1 s.d.** or **z-score**
- For each variable x_i (excluding x_0):
 - replace with $(x_i - \mu_i) / \sigma_i$
- Reversible operation (may not be relevant)

0 Mean Data			
x_0	$x_1 - \mu_1$	$(x_1 - \mu_1)^2$	y
1	-25	625	1.8
1	-13	169	2.1
1	3	9	6.1
1	35	1225	7.8
Sum	0	2028	
Mean	0	507 $\sigma_1 = 22.51$	

0 Mean, 1 s.d. Data			
x_0	$x_1 - \mu_1$	$(x_1 - \mu_1) / \sigma_1$	$(x_1 - \mu_1)^2 / \sigma_1^2$
1	-25	-1.11	1.232
1	-13	-0.577	0.333
1	3	0.133	0.0177
1	35	1.554	2.414
Sum	0	0	4
Mean	0	0	1 $\sigma_1 = 1$

Z scores = location + scale transform

- Z scores refers to data points that are mapped to 0 mean, 1 s.d. *normal* distribution
- Transform operations can be viewed as:
instead of learning $f(\mathbf{X})=\mathbf{y}$ learn $f((\mathbf{X} - \mu)/\sigma)=\mathbf{y}$
- For regression problems if the output values (i.e. the $\mathbf{y}$ values) are also transformed then they will need to be mapped back

When to apply Data standardisation?

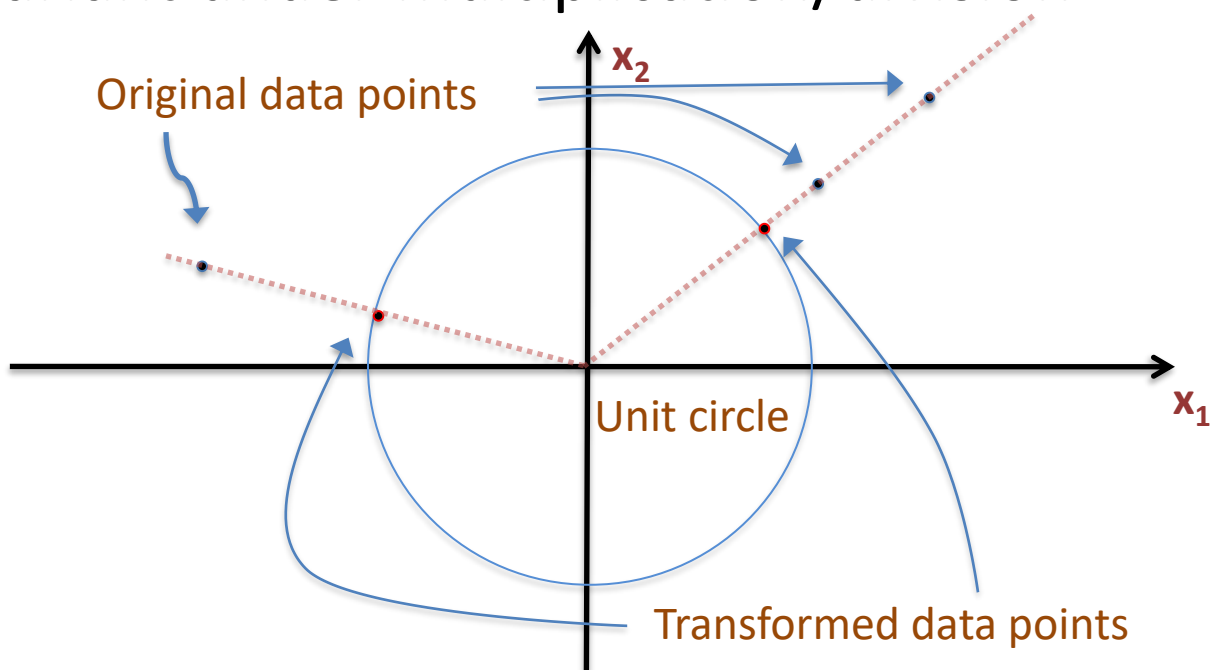
- Data standardisation needs to be handled with care.
- When higher order polynomials are involved, data standardisation will cause higher order variables with much larger ranges scaled to the same range.
- This will result in the errors to be scaled thus larger errors associated with higher order variables will be scaled to smaller values.
- Experiment with it before applying!

Normed Data

- For some applications (e.g. Text classification, Clustering) the input data often is *normed*
- **Normed** data assumes that each input data point is a unit vector in Euclidean space
- Given $\mathbf{x} = \langle x_1, x_2, \dots, x_k \rangle$
- The squared/Euclidean norm $|\mathbf{x}| = \sqrt{x_1^2 + x_2^2 + \dots + x_k^2}$
- The **Normed** version of $\mathbf{x}$ can be generated by applying the transformation:

$$\langle x_1, x_2, \dots, x_k \rangle \mapsto \left\langle \frac{x_1}{|\mathbf{x}|}, \frac{x_2}{|\mathbf{x}|}, \dots, \frac{x_k}{|\mathbf{x}|} \right\rangle$$

- Maps data points into the unit (hyper) sphere
- **Retains the angle** between the original vectors
- But **not the (Euclidean) distance**
- Invariant under multiplication/division



- For regression problems, normed data would not be appropriate as many data points will have the same normed representation.
- e.g. For $\langle \text{age, weight} \rangle$: $\langle 2, 4 \rangle$ and $\langle 20, 40 \rangle$
if we are predicting **height**, normed transformation would imply the height would be the same in these two cases
- **Note** that norm transform is not reversible

Geometrical perspective

- All data transformation methods including basis expansion
- can be seen as transformation from the original multidimensional space to a different multidimensional space.
- Norm transformation projects all datapoints into the unit (hyper) sphere
- Can be understood even more generally in terms of **Kernel spaces**